Heat Transfer to a Fluid in a Stirred Tank

Group # 4

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Chemical Engineering Lab II
02/04/2020

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Introduction

- **Objective**
  - Calculate the overall heat transfer coefficient (U) range for a variety of heat transfer processes by varying the flow rate of cooling water from the jacket around the vessel, the speed of the impeller inside the vessel and by adding/removing the baffle for a steady state.
- **Technical nature**
  - Setting up the apparatus to obtain steady state conditions
  - Controlling the impeller speed, by varying it to 600,750,900 RPM
  - Controlling the cooling water flowrate
  - Obtaining readings from the temperature gauges
  - Collecting the vapor condensate from the cooling coil

Theory

- Heat$_{(in)}$ = Heat$_{(out)}$ for the steady state condition heat balance
- At steady state, the heat transferred Q is the same at any point in the x direction
- Different interface between solids phases will represent a different resistance of heat transfer, which will result in a temperature profile, represented by the equation

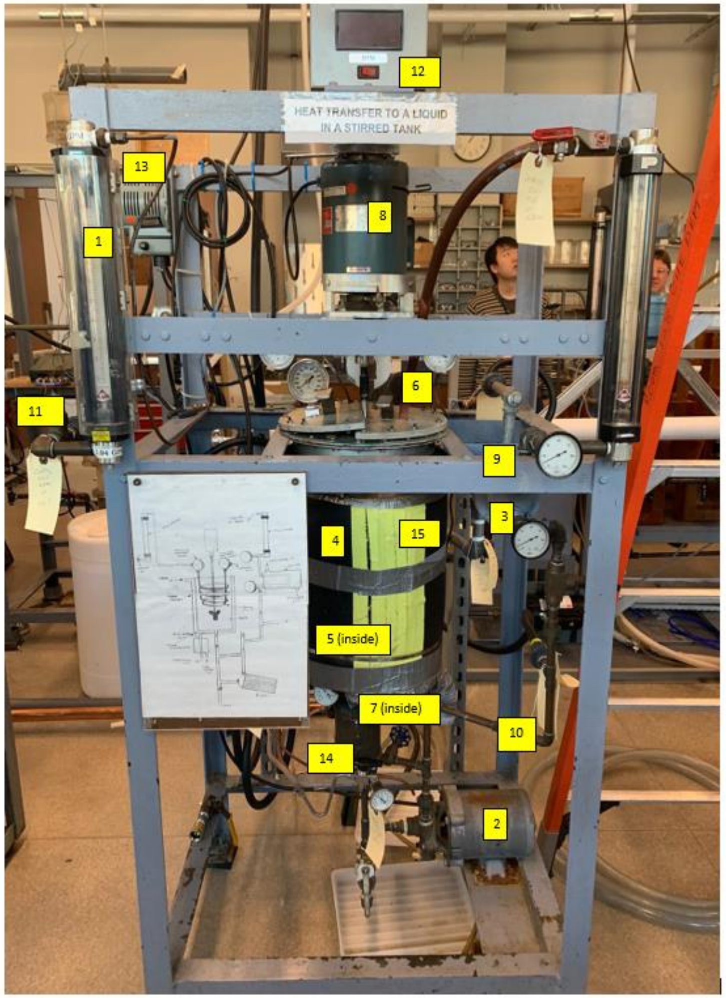
$$Q = h_i A_i (t_4 - t_5) = h_o A_o (t_1 - t_2)$$
$$Q_{13} = Q_{hx} + Q_c$$

  - Q_{13} is the heat transferred from the steam to the vessel $\left[\frac{J}{s}\right]$
  - Q_{hx} and Q_c are the heat removed by the heat exchanger and the cooling coil respectively $\left[\frac{J}{s}\right]$
- Assuming the wall thickness is small compared to the tank diameter:

$$U_i = \frac{1}{\frac{1}{h_i} + \frac{1}{h_0}} \qquad Q_{13} = U_{i,wall}(t_1 - t_3)A$$

  - U is the overall heat transfer coefficient $\left[\frac{W}{m^2 K}\right]$
  - A is the area of heat transfer $[m^2]$
  - t_1 and t_3 are the steam and vessel temperatures [K]

Apparatus



Serial No.	Apparatus	Description
1.	flowmeter (2)	Device to measure the fluid flowrate inside the system
2.	Pump	Mechanical device use to push fluid through the piping system
3.	Cooler Tank	Jacketed vessel with recirculation of steam and water through a cooling coil
4.	Steam Jacket	Enclosure surrounding the tank, which allows the steam to flow
5.	Cooling Coil	Coil tubing inside the tank, which allows the cold water to flow
6.	Thermometers	Device that measures the temperature of the flowing fluid at different locations inside the unit.
7.	Impeller	Agitator that stirs the liquid inside the tank, ensures a uniform mixing. It can be adjusted for higher speeds
8.	Motor	Mechanical device that drives the impeller inside the tank by supplying power
9.	Heat Exchanger	Shell and tube heat exchanger type, which allows the transfer of heat between a cold and hot stream
10.	Piping	Tubing system which allows fluid to be transported in the unit
11.	Valves	Device that controls the fluid flow inside the piping
12.	Tachometer	Device use to measure the RPM
13.	Variac	Device use to set the voltage delivered to the system
14.	Stream Trap	Use to contain the steam coming out of the steam jacket
15.	Insulation	Material surrounding the tank sue to minimize heat loss or gain from the surrounding

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Material & Supplies

Serial Number	Material	Description
1.	Measuring scale	For measuring the size of the tank
2.	Water	Water used to fill the tank and used as cooling water in the heat exchanger and condensate steam
3.	Steam	Used as a hot stream of steam through the steam jacket to heat the water in the unit
4.	Baffles	Used to promote mixing by creating turbulence and increase heat transfer in the tank
5.	Stopwatch	Device use to record time during collection of fluid or condensate
6.	Graduate Cylinder	Device used to measure the volume of fluid collected

Procedure

- Fill the tank with water until 2 in from the top
- Make sure all the valves are in the right position
- Turn on the impeller and adjust the speed to 600 RPM
- Carefully open the valve for the cooling water for the coils
- Turn on cooling water for the heater exchanger
- Make sure the suction and bypass valves are fully open while the outlet valve is fully closed
- Turn on the pump and let it run for a few seconds with the discharge valve closed
- Open the discharge valve slowly
- Open the inlet steam valve ¼ of a turn

Procedure

Experiment A

- Set the inlet steam so the vessel is kept between 65 and 80℃
- Record all the temperatures in the gauges
- Repeat prior steps for 750 and 900RPM at steady state
- Repeat prior steps for 750RPM, however now use a different flowrate for the heat exchanger
- Repeat last step twice for different flow

Experiment B

- Repeat experiment A with the baffles.

Experimental Challenges

- Determining the valves & order described by the manual
- Setting up the steam flowrate in order to obtain the desired temperature
- Reading of the flowrate for the coil cooling water (fluctuates a lot)
- Setting the heat exchanger cooling water flowrate constant (valve is leaking)
- Putting the baffles back in the vessel & waiting for steady state

Safety

- Wear eyes glasses
- Wear thermal gloves when installing baffles inside the vessel or removing them when equipment is hot
- Steam pipeline is very hot, be careful with accidentally contact
- Turn off the impeller when installing/removing baffles
- Be cautious when placing baffles to the heat exchanger pipe viscidity
- Turning RPM too high where water splashes
- Over-filling the tank where spills may cause a slippery floor